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OOPS (PPF)

Mini-Project Assignment

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Shape Area & Perimeter Calculator

Problem Statement

Develop a calculator that computes the area and perimeter of different shapes through a common interface.

Class Design

Shape (Base Class)
|

| | |
Circle Rectangle Triangle

- Shape is an abstract base class that declares the virtual area() and perimeter() functions.
- Circle, Rectangle, and Triangle inherit from Shape.
- Each derived class implements its own formulas.
- The Shape* pointer calls the appropriate functions at runtime

Features

- Calculates the area and perimeter of a circle.
- Calculates the area and perimeter of a rectangle.
- Calculates the area and perimeter of a triangle.
- Uses a menu-driven console interface.
- Checks for invalid dimensions.
- Allows users to perform multiple calculations until they choose to exit.

OOPS CONCEPTS USED

1. **Classes and Objects:** Each shape is represented using a class.
2. **Inheritance:** Circle, Rectangle, and Triangle inherit from the Shape class.
3. **Abstraction:** The Shape class defines common pure virtual functions for calculating area and perimeter, becoming an abstract class.
4. **Virtual Functions:** The `area()` and `perimeter()` functions are declared as virtual functions.
5. **Function Overriding:** Each derived class provides its own implementation of the functions.
6. **Runtime Polymorphism:** A base-class pointer is used to call the appropriate functions for each shape.
7. **Encapsulation:** Shape dimensions are stored as private data members.

Formulas Used

Shape	Area	Perimeter
Circle	πr^2	$2\pi r$
Rectangle	$l \times w$	$2(l + w)$
Triangle	$\sqrt{s(s-a)(s-b)(s-c)}$	$a + b + c$

Invalid Inputs

- Menu Choices outside 1-4 are rejected.
- Input Values (for side/length/width/radius) must be greater than 0.
- Input values must be numeric.
- A triangle sides must satisfy that sum of any two sides always greater than third side for them to be valid.

Test Cases

1. Circle Output

```
===== SHAPE CALCULATOR =====
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice: 1
Enter radius: 5
Area = 78.5397
Perimeter = 31.4159
```

2. **Rectangle Output**

```
===== SHAPE CALCULATOR =====
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice: 2
Enter length: 12
Enter width: 10
Area = 120
Perimeter = 44
```

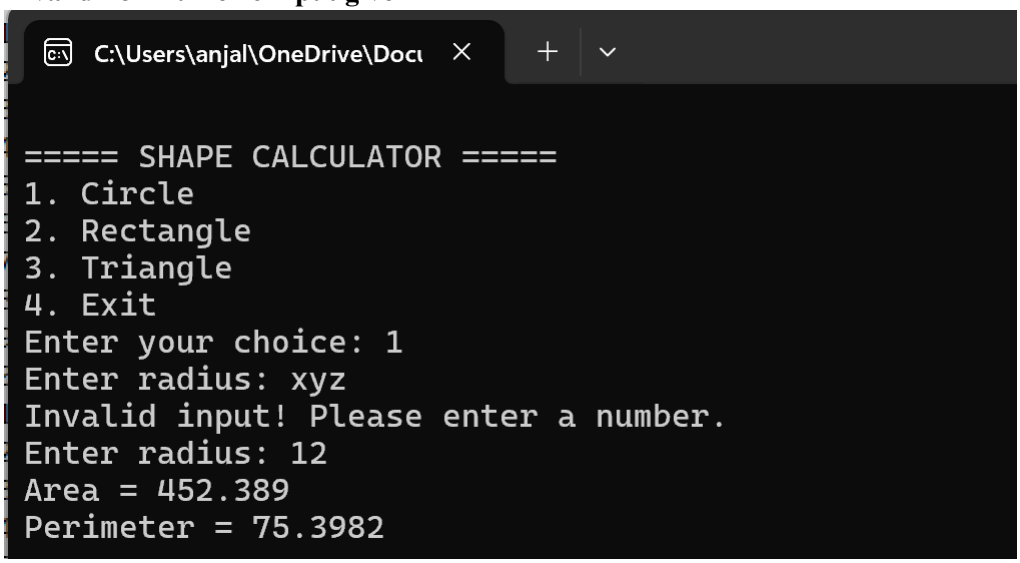
3. **Triangle Output**

```
===== SHAPE CALCULATOR =====
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice: 3
Enter first side: 3
Enter second side: 4
Enter third side: 5
Area = 6
Perimeter = 12
```

4. **Invalid Choice Given**

```
===== SHAPE CALCULATOR =====
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice: 10
Invalid choice. Please select 1 to 4.
```

5. **Invalid non-numeric input given**



6. **Invalid negative sides given**

```
===== SHAPE CALCULATOR =====
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice: 2
Enter length: -1
Enter width: 5
Length and width must be positive.
```

7. **Invalid triangle sides (sum of two sides not greater than or equal to third side)**

```
===== SHAPE CALCULATOR =====
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice: 3
Enter first side: 1.234
Enter second side: 2
Enter third side: 34
Invalid triangle sides.
```

Conclusion

The Shape Area & Perimeter Calculator calculates the area and perimeter of circles, rectangles, and triangles using mathematical formulas. It uses OOPS concepts such as classes and objects, inheritance, encapsulation, abstraction, virtual functions, function overriding, and runtime polymorphism. The program also validates user inputs and handles invalid dimensions to ensure accurate calculations. Its menu-driven interface makes the program simple, organized, and user-friendly.